

2.5.2 Non-Linear Data Structures: Tree

ٹری ڈیٹا کو درجہ بندی (Hierarchy) میں رکھتی ہے جو ایک جڑ (Root) سے پھوٹتی ہے۔

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 19 of 24

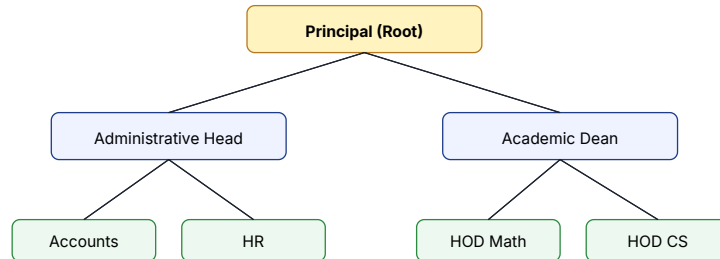
2.5.2 Non-Linear Data Structures: Tree

LEARN IT

A **tree** organises data in a **hierarchy**. Everything branches out from a single starting point called the **root**.

It creates a strict **one-to-many (parent-child)** relationship. That is an excellent way to show how roles flow in an organisation.

Real-life examples: a family tree (grandparents → parents → children) and folders inside folders on a drive.



College tree: one root (Principal), then parent-child branches. No child has two parents in this model.

WORKED EXAMPLE

Worked example: name the relationships.

- Root: Principal.
- Children of the root: Administrative Head, Academic Dean.
- Children of Academic Dean: HOD Math, HOD CS.
- HOD CS has one parent (Academic Dean), not two — that is the tree rule. If HOD CS also reported sideways to Accounts as an equal link, the picture would become a **graph**.

REMEMBER IT

Tree = family / folders / organogram. One root. Branches down. Parent-child, not a web of equals.

COMMON MISTAKE

Mistake: drawing a tree with two roots, or with an arrow going back up to the Principal from HOD CS (a cycle).

Fix: one root, links only parent → child, no loops. Cycles belong to graphs.

اردو میں سمجھیں

ٹری ڈیٹا کو درجہ بندی (Hierarchy) میں رکھتی ہے جو ایک جڑ (Root) سے پھوٹی ہے۔ تعلق والدین-بچہ (ایک سے کئی) ہوتا ہے، جیسے پرنسپل → ڈین → ایچ او ڈی، خاندانی شجرہ، یا فولڈر کے اندر فولڈر۔ دو جڑیں یا چکر ٹری کو گراف بنا دیتے ہیں۔

CHECK YOURSELF

1. What is the starting node of a tree called?
2. What kind of relationship does a tree model?
3. Give two real-life tree examples from the source.

Answers: 1. The root. 2. One-to-many parent-child. 3. A family tree, and folders inside folders (or a college organogram).

Next lecture: Non-Linear Data Structures: Graph